

AC Motor 365 DAYS PM CHECKLIST

For standard 480V motor

EQ#: _____ Unit: _____ SN# _____ WO# _____ Date: _____

PM Cycle: 365 Days

EQ	PM Item	Standard	Result	Comment						
Motor	1 Winding Insulation Resistance Measurement	Perform Insulation resistance testing if the motor is offline ☐ Yes (Motor Offline) ☐ No (Motor Online) 480V Motor: Megger 1.5MΩ (1KV test 60s)	PASS ☐ Abnormal ☐ N/A ☐ Phase-Ground (MΩ): <table border="1"> <tr> <td>A-G</td><td>B-G</td><td>C-G</td></tr> <tr> <td></td><td></td><td></td></tr> </table>	A-G	B-G	C-G				
A-G	B-G	C-G								
2 Three-Phase Winding Resistance Measurement	Perform winding resistance testing if the motor is offline ☐ Yes (Motor Offline) ☐ No (Motor Online) Phase resistance deviation within ±3%	PASS ☐ Abnormal ☐ N/A ☐ Phase-Phase (Ω): <table border="1"> <tr> <td>A-B</td><td>B-C</td><td>C-A</td></tr> <tr> <td></td><td></td><td></td></tr> </table>	A-B	B-C	C-A					
A-B	B-C	C-A								
3 Appearance	No rust, damage or contamination; nameplate legible	PASS ☐ Abnormal ☐								
4 Grounding	Grounding conductor (motor frame) firmly connected; no looseness; grounding resistance < 5 Ω	PASS ☐ Abnormal ☐ _____ Ω								
5 Accessories	No abnormalities; indicators function properly	PASS ☐ Abnormal ☐								
6 Push Button Station	No rust, damage or contamination; Function properly	PASS ☐ Abnormal ☐ N/A ☐								
7 Terminal Box & Conduit	Connection secure; conduit in good condition; no water	PASS ☐ Abnormal ☐ N/A ☐								
8 Terminal Box Internal Wiring (From motor)	No discoloration or deterioration	PASS ☐ Abnormal ☐ N/A ☐								
9 Terminal Box Terminal Conductors (From outside)	No discoloration or deterioration; fixed securely; no stress damage	PASS ☐ Abnormal ☐ N/A ☐								
10 Terminal Box Internal Terminal	Firmly fixed; no abnormalities	PASS ☐ Abnormal ☐ N/A ☐								
11 Terminal Box Internal Terminal Insulation	Firmly secured; no looseness or damage	PASS ☐ Abnormal ☐ N/A ☐								
12 Terminal Box Internal Ground Wires	Insulation intact; no looseness; grounding normal	PASS ☐ Abnormal ☐ N/A ☐								
13 Terminal Box Internal Environment	Dry and clean; no moisture	PASS ☐ Abnormal ☐ N/A ☐								
Lubrication System	1 Automatic Grease Feeder	Lubrication piping unobstructed	PASS ☐ Abnormal ☐ N/A ☐							
	2 Grease Refill	New grease must reach outlet after refill	PASS ☐ Abnormal ☐							

	3	Lubricating Oil Replacement	Oil level at half gauge (1/2)	PASS ☐ Abnormal ☐ N/A ☐	
Cooling System	1	Cooling Fan	Fan grille clean and unobstructed	PASS ☐ Abnormal ☐	
	2	Cooling System Check	No rust; heat exchanger and duct clean and undamaged	PASS ☐ Abnormal ☒	
MCC or Starter	1	Breaker	Breaker Rating	_____ A	
			Current Limiter	_____ A	
			Trip Unit	PASS ☐ Abnormal ☐ N/A ☐	
			Current Limiter	PASS ☐ Abnormal ☐ N/A ☐	
			Wire Connection	PASS ☐ Abnormal ☐ N/A ☐	
	2	Contactor	Size	_____	
			Type	VFD ☐ R ☐ 1S ☐ 2S ☐	
			Main Contact	PASS ☐ Abnormal ☐ N/A ☐	
			Coil / Core	PASS ☐ Abnormal ☐ N/A ☐	
			Arc Chute	PASS ☐ Abnormal ☐ N/A ☐	
			Wire Connection	PASS ☐ Abnormal ☐ N/A ☐	
	3	Overload Relay	Heater Size	_____	
			Heaters	PASS ☐ Abnormal ☐ N/A ☐	
			Reset Mechanism	PASS ☐ Abnormal ☐ N/A ☐	
			Wire Connection	PASS ☐ Abnormal ☐ N/A ☐	
	4	Others	Protection Relay	PASS ☐ Abnormal ☐ N/A ☐	
			Auxiliary Relay	PASS ☐ Abnormal ☐ N/A ☐	
			CPT	PASS ☐ Abnormal ☐ N/A ☐	
			CT	PASS ☐ Abnormal ☐ N/A ☐	
			Fuses	PASS ☐ Abnormal ☐ N/A ☐	
			Push Buttons	PASS ☐ Abnormal ☐ N/A ☐	
			Indicating Lights	PASS ☐ Abnormal ☐ N/A ☐	
			Terminal Blocks	PASS ☐ Abnormal ☐ N/A ☐	
			Control Wiring	PASS ☐ Abnormal ☐ N/A ☐	
			Wire Raceway	PASS ☐ Abnormal ☐ N/A ☐	
			Handle Mechanism	PASS ☐ Abnormal ☐ N/A ☐	

			Discoloration	PASS ☐ Abnormal ☐ N/A ☐																						
			Soft-Start	PASS ☐ Abnormal ☐ N/A ☐																						
			Mechanical Interlock	PASS ☐ Abnormal ☐ N/A ☐																						
			Auto Restart Function Check	PASS ☐ Abnormal ☐ N/A ☐																						
			Motor Starter Cubicle Cleaness	PASS ☐ Abnormal ☐ N/A ☐																						
	5	Infrared Survey (IR)	Motor standby or starter not accessible	Yes ☐ No ☐																						
			Motor Condition: ☐ Standby ☐ Running Current (A):	<table border="1"> <thead> <tr> <th>A</th> <th>B</th> <th>C</th> </tr> </thead> <tbody> <tr> <td> </td> <td> </td> <td> </td> </tr> </tbody> </table> L1 L2 L3 T1 T2 T3 TEMPERATURE (°F) AØ BØ CØ <table border="1"> <tr> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> </tr> </table>		A	B	C																		
			A			B	C																			
		PASS ☐ Abnormal ☐ N/A ☐																								
Complete PM	1	Complete work	Close the cabinet and door	Yes ☐ No ☐																						
	2	THM check	lean the environment when complete the PM	Yes ☐ No ☐																						

Inspected By:_____

AC Motor 365 DAYS PM CHECKLIST

For standard 5kV&15kV motor

EQ#: _____ Unit: _____ SN# _____ WO# _____ Date: _____ PM Cycle: 365 Days

EQ	PM Item	Standard	Result	Comment						
Motor	1	Winding Insulation Resistance Measurement Perform Insulation resistance testing if the motor is offline ☐ Yes (Motor Offline) ☐ No (Motor Online) Insulation resistance test 5kV Motor: Megger 5MΩ (2.5KV test 60s) 15kV Motor: Megger 12MΩ (5KV test 60s)	PASS ☐ Abnormal ☐ N/A ☐ Phase-Ground (MΩ): <table border="1"> <tr> <td>A-G</td> <td>B-G</td> <td>C-G</td> </tr> <tr> <td></td> <td></td> <td></td> </tr> </table>	A-G	B-G	C-G				
	A-G	B-G	C-G							
	2	Dielectric Absorption Ratio, Polarization Index (P.I.) Measurement Absorption ratio R60/R30 > 1.25 Polarization Index P.I. = R10/R1 > 2	PASS ☐ Abnormal ☐ N/A ☐							
	3	Three-Phase Winding Resistance Measurement Perform winding resistance testing if the motor is offline ☐ Yes (Motor Offline) ☐ No (Motor Online) Phase resistance deviation within ±3%	PASS ☐ Abnormal ☐ N/A ☐ Phase-Phase (Ω): <table border="1"> <tr> <td>A-B</td> <td>B-C</td> <td>C-A</td> </tr> <tr> <td></td> <td></td> <td></td> </tr> </table>	A-B	B-C	C-A				
	A-B	B-C	C-A							
	4	Appearance No rust, damage or contamination; nameplate legible	PASS ☐ Abnormal ☐							
	5	Grounding Grounding conductor (motor frame) firmly connected; no looseness; grounding resistance < 5 Ω	PASS ☐ Abnormal ☐ _____ Ω							
	6	Accessories No abnormalities; indicators function properly	PASS ☐ Abnormal ☐							
	7	Push Button Station No rust, damage or contamination; Function properly	PASS ☐ Abnormal ☐ N/A ☐							
	8	Terminal Box & Conduit Connection secure; conduit in good condition; no water	PASS ☐ Abnormal ☐ N/A ☐							
	9	Terminal Box Internal Wiring (From motor) No discoloration or deterioration	PASS ☐ Abnormal ☐ N/A ☐							
10	Terminal Box Terminal Conductors (From outside) No discoloration or deterioration; fixed securely; no stress damage	PASS ☐ Abnormal ☐ N/A ☐								
11	Terminal Box Internal Terminal Firmly fixed; no abnormalities	PASS ☐ Abnormal ☐ N/A ☐								
12	Terminal Box Internal Terminal Insulation Firmly secured; no looseness or damage	PASS ☐ Abnormal ☐ N/A ☐								
13	Terminal Box Internal Ground Wires Insulation intact; no looseness; grounding normal	PASS ☐ Abnormal ☐ N/A ☐								

	14	Terminal Box Internal Environment	Dry and clean; no moisture	PASS ☐ Abnormal ☐ N/A ☐					
Lubrication System	1	Automatic Grease Feeder	Lubrication piping unobstructed	PASS ☐ Abnormal ☐ N/A ☐					
	2	Grease Refill	New grease must reach outlet after refill	PASS ☐ Abnormal ☐					
	3	Lubricating Oil Replacement	Oil level at half gauge (1/2)	PASS ☐ Abnormal ☐ N/A ☐					
Cooling System	1	Cooling Fan	Fan grille clean and unobstructed	PASS ☐ Abnormal ☐					
	2	Cooling System Check	No rust; heat exchanger and duct clean and undamaged	PASS ☐ Abnormal ☐					
MCC or Starter	1	Breaker	SWGR No#	_____					
			Breaker Rating	_____ A					
			Setting	_____ A					
			Running Amps	<table border="1"> <tr> <td>A</td><td>B</td><td>C</td></tr> <tr> <td></td><td></td><td></td></tr> </table>	A	B	C		
	A	B	C						
	2	MCC/Starter Inspection	Power Fuse	PASS ☐ Abnormal ☐ N/A ☐					
			CPT Fuse	PASS ☐ Abnormal ☐ N/A ☐					
			Insulators	PASS ☐ Abnormal ☐ N/A ☐					
			Phase Barriers	PASS ☐ Abnormal ☐ N/A ☐					
			Vacuum Interrupters	PASS ☐ Abnormal ☐ N/A ☐					
			Magnetic Coil	PASS ☐ Abnormal ☐ N/A ☐					
			Protection Relays	PASS ☐ Abnormal ☐ N/A ☐					
			Control Transformer	PASS ☐ Abnormal ☐ N/A ☐					
			Wire Connection	PASS ☐ Abnormal ☐ N/A ☐					
			Push Buttons	PASS ☐ Abnormal ☐ N/A ☐					
			Indication Lights	PASS ☐ Abnormal ☐ N/A ☐					
Mechanical Interlock			PASS ☐ Abnormal ☐ N/A ☐						
VFD			PASS ☐ Abnormal ☐ N/A ☐						
Complete PM	1	Complete work	Close the cabinet and door	Yes ☐ No ☐					
	2	THM check	Clean the environment when complete the PM	Yes ☐ No ☐					

Reviewed By: _____

Inspected By: _____

AC Motor 365 DAYS PM CHECKLIST

For all critical motor
annual visual inspection

EQ#: _____ Unit: _____ SN# _____ WO# _____ Date: _____ PM Cycle: 365 Days

EQ	PM Item	Standard	Result	Comment
Motor	1 Appearance	No rust, damage or contamination; nameplate legible.	PASS ☐ Abnormal ☐	
	2 Guards for Rotating Parts	No looseness, no damage, no missing guards	PASS ☐ Abnormal ☐	
	3 Grounding	1. Grounding wire connect to ground terminal on the motor frame, cannot connect to terminal box, fan cover or foundation bolt. 2. Grounding wire firmly connected, no looseness. 3. Grounding resistance < 5 Ω	PASS ☐ Abnormal ☐ _____ Ω	
	4 Accessories	No damage; indicators show no abnormalities	PASS ☐ Abnormal ☐	
	5 Push Button Station	No rust, damage or contamination; Function properly	PASS ☐ Abnormal ☐ N/A ☐	
	6 Terminal Box & Conduit	No looseness or damage; no water or dust ingress; check T-leads	PASS ☐ Abnormal ☐ N/A ☐	
	7 Bearing Temperature	No abnormal high temperature (for units equipped with temperature indicator)	PASS ☐ Abnormal ☐ N/A ☐	
	8 Space Heater	Perform Ohm test	PASS ☐ Abnormal ☐ N/A ☐	
	9 Explosion-Proof Rating Check	1. Body undamaged and not deformed. 2. Conduits and cable terminations tight and undamaged; sealing compound intact. 3. Explosion-Proof related parts meet requirements and working normally.	PASS ☐ Abnormal ☐ N/A ☐	
Lubrication System	1 Automatic Grease Feeder	Operating normally.	PASS ☐ Abnormal ☐ N/A ☐	
	2 Grease Outlet	Grease discharge normal at outlet; waste-grease collector working properly.	PASS ☐ Abnormal ☐	
	3 Lubricating Oil Level	Oil level at mid-gauge (about 1/2).	PASS ☐ Abnormal ☐ N/A ☐	
	4 Lubricating Oil Pressure	_____ PSI	PASS ☐ Abnormal ☐ N/A ☐	
	5 Lubricating Oil Temperature	_____ °F	PASS ☐ Abnormal ☐ N/A ☐	
	6 Loaded-End Bearing Oil Temperature	Normal.	PASS ☐ Abnormal ☐ N/A ☐	

	7	Non-Loaded-End Bearing Oil Temperature	Normal.	PASS ☐ Abnormal ☐ N/A ☐	
Cooling System	1	Cooling Fan	Fan grille clean, unobstructed	PASS ☐ Abnormal ☐	
MISC	1	Abnormal Odor	No odor	PASS ☐ Abnormal ☐	
	2	Abnormal Noise	No abnormal noise	PASS ☐ Abnormal ☐	
	3	Abnormal Vibration	No abnormal vibration	PASS ☐ Abnormal ☐	
	4	Push Button Station	No rust, no damage, no contamination, nameplate legible.	PASS ☐ Abnormal ☐	
	5	Motor Body Temperature	Normal temperature, no overheating (for units with temperature indicator)	PASS ☐ Abnormal ☐	
Complete PM	1	THM check	Clean the environment when complete the PM	Yes ☐ No ☐	

Reviewed By:_____ **Inspected By:**_____